Higher-Order Topological Deep Learning for Predicting Human Review Effort in AI-Generated GitHub Pull Requests

CSC 7333: Machine Learning | Group Project Proposal

Group members: Arif Ali, Abdulla M Asad, Chloe G Kinsley, Shanice Manning, Egan M Trahant, Nabila Fairuz.

Research question. Can representing GitHub development workflows as combinatorial complexes improve the prediction of future human review effort relative to models using PR-level features or pairwise graph relations?

1. Problem Statement

AI coding agents such as GitHub Copilot, Claude Code, OpenAI Codex, Cursor, and Devin can generate code and submit pull requests (PRs) to GitHub repositories. Their contributions still require human review, and some PRs lead to substantial discussion or repeated revisions. Predicting which PRs are likely to need more review could help maintainers plan their work.

Most prediction methods use PR-level statistics or pairwise links between software-development entities. These representations do not always distinguish the multiway events that occur during development: for example, a particular commit modifying a particular file within a PR, or a reviewer submitting a particular review of that PR. Our project examines whether retaining these joint relationships is useful for prediction.

2. Project Objective

We will build a hierarchy-aware topological model to classify AI-authored PRs by their *future human review effort*. The main comparison is between a model that preserves higher-order relationships and baselines trained on tabular features or ordinary graphs.

Our starting point is the use of combinatorial complexes in *Beyond Flat Models: Preserving ICS Topology using Combinatorial Complexes for Hierarchy-Aware Intrusion Detection*. We apply the same general idea of retaining meaningful group relationships to a different setting: GitHub code changes and review activity.

3. Dataset

We plan to use *AIDev: Studying AI Coding Agents on GitHub*, focusing on the AIDev-pop subset, which contains linked PR, commit, file-change, and review records. The full AIDev v5 release includes approximately 7.69 million AI-authored PRs; a smaller sample of the detailed subset will be used for initial experiments.

Data source	Role in the project
Pull requests and repositories	PR identification, agent attribution, repository context, and timestamps
Commits and modified files	PR–commit–file relationships and code-change summaries
Reviews and review comments	Review participants, observed review events, and future-effort labels

4. Higher-Order Topological Representation

We will represent the GitHub workflow as a heterogeneous rank-2 combinatorial complex $C = (V, \mathcal{X}, \mathrm{rk})$. Here, V contains the GitHub entities, $\mathcal{X}$ contains the cells, and $\mathrm{rk} \in \{0, 1, 2\}$ specifies a cell's rank. Rather than converting all activity into an untyped graph, we retain both entity types and the groups in which they participate.

Rank	Interpretation	Examples
0-cells	Individual entities	AI agent, developer, PR, commit, file, review, repository
1-cells	Pairwise relations	PR–commit, commit–file, PR–review, review–human
2-cells	Multiway interactions	(PR, commit, modified file) and (PR, review, reviewer)



Figure 1: Illustrative 2-cells constructed from observed code-change and review events.

A **code-change 2-cell** connects a PR, one of its commits, and a file changed by that commit. A **review 2-cell** connects a PR, an individual review event, and its author. We will also test larger PR-centered groups as an alternative construction. The diagrams show three-member cells, but combinatorial-complex 2-cells need not be limited to three entities.

Incidence matrices B_{01} and B_{12} record which 0-cells belong to 1-cells and which 1-cells belong to 2-cells. This lets the model exchange information across ranks. A combinatorial complex does not require every possible pairwise face of a higher-order cell to be present.

5. Prediction Task

We will make predictions **24 hours after PR creation**, using AI-authored PRs that are still open at that time and have an observable follow-up period.

Component	Definition
Input	PR metadata and relationships verified to exist within the first 24 hours
Prediction window	The seven days following the 24-hour observation point
Review effort	Number of human-submitted review events plus inline review comments in that window
Binary target	High effort if the count exceeds a threshold fitted to the training cohort (approximately the top 20%); low effort otherwise

Prediction target: $\Pr\big(Y_p = 1 \mid \text{GitHub activity observed through hour } 24\big)$.

The threshold will be fixed using the training data only. Relations without reliable observation times will be left out of historical snapshots rather than treated as if they were known at prediction time.

6. Proposed Methodology

We will implement the system in five stages.

- **Prepare the data.** Join AIDev records with stable identifiers, remove duplicate entries, distinguish human users from bots, and construct time-valid PR snapshots.
- **Build the complex.** Create the 0-, 1-, and 2-cells from verified relationships. Measure how often each cell type occurs and how much cells overlap.
- **Learn representations.** Train a Topological Neural Network (TNN), specifically a Combinatorial Complex Neural Network (CCNN), with message passing between adjacent cell ranks.
- **Predict review effort.** Combine PR-level embeddings with a classifier trained to distinguish high from low future review effort.
- **Analyze results.** Compare models, inspect failures, and test the effects of different 2-cell definitions.

7. Baseline Models and Validation

We will compare models using the same information available at prediction time. The goal is to isolate the effect of the representation, not to give one model more features or access to later events.

Model	Representation / role
Logistic Regression; LightGBM	PR-level statistics and collaboration summaries
GraphSAGE or Graph Attention Network	Pairwise entity relations; graph-learning baseline
Hypergraph Neural Network	Multiway groups without explicit cell ranks
Combinatorial Complex Neural Network	Proposed rank-0/1/2 model using incidence relations

Performance measures. We will report AUROC, average precision, precision, recall, F1-score, and calibration, together with training time and memory use. Class balance and variability across repeated runs will also be recorded.

Generalization. A chronological split will test prediction on later PRs. A separate repository-disjoint split will test performance on repositories excluded from training. Preprocessing, label threshold selection, and graph construction will be fitted or restricted to the relevant training partition and prediction cutoffs.

Ablation tests. We will compare a model using only 0- and 1-cells with models that add code-change 2-cells, review 2-cells, or PR-centered groups. We will also examine a control in which 2-cell memberships are shuffled while approximately preserving cell sizes and participation counts.

Feasibility check. Before training, we will count eligible PRs, valid reviews, timestamped file-change observations, and usable multiway cells. If coverage is insufficient at the proposed 24-hour cutoff, we will simplify the inputs or reconsider the observation window without introducing future-data leakage.

8. Research Questions

ID	Research question
RQ1	Does AIDev contain enough reliable higher-order relationships to support combinatorial-complex learning?
RQ2	Does the proposed topological model improve review-effort prediction over tabular and pairwise graph baselines?
RQ3	Which higher-order groups are most useful: code-change interactions, review interactions, or PR-centered groups?
RQ4	Does higher-order structure help a model generalize to repositories it has not seen during training?

9. Expected Contributions

- A reproducible procedure for constructing and checking a combinatorial complex from AI-authored GitHub development records.
- A working model for predicting human review effort, evaluated against tabular, graph, and hypergraph baselines.
- Ablation and generalization experiments showing when multiway relations do, or do not, add predictive information.
- An analysis of the development and review interactions associated with higher review effort.

10. Expected Outcome and Scope

We want to determine whether the structure of a development workflow offers useful predictive information beyond familiar PR statistics and ordinary graphs. Improved performance is a hypothesis, not an assumption. A result showing similar accuracy with greater model complexity would also be informative, particularly if it identifies which relations contribute little.

For the course project, we will prioritize a well-checked data sample, reliable baselines, one functioning higher-order model, and a small set of controlled experiments. Broader architecture searches and large-scale interpretability studies would be follow-up work if the initial results warrant them.

References and Implementation Resources

1. Li et al., *AIDev: Studying AI Coding Agents on GitHub*. `https://arxiv.org/abs/2602.09185`.
2. *Beyond Flat Models: Preserving ICS Topology using Combinatorial Complexes for Hierarchy-Aware Intrusion Detection*. IEEE. `https://ieeexplore.ieee.org/document/11754058`.
3. *Early-Stage Prediction of Review Effort in AI-Generated Pull Requests*. `https://arxiv.org/abs/2601.00753`.
4. AIDev dataset, Hugging Face. `https://huggingface.co/datasets/hao-li/AIDev`.

Implementation: Python, PyTorch, scikit-learn, LightGBM, TopoNetX, and TopoModelX; experiments in Google Colab or another suitable computing environment.